Solution to Problem 1.

(a) The mutual coherence is defined as

$$\mu(\boldsymbol{A}) = \max_{i \neq j} |\langle \boldsymbol{a}_i, \boldsymbol{a}_j \rangle| = \max_{i \neq j} \left| \boldsymbol{a}_i^T \boldsymbol{a}_j \right|.$$

[2]

(b)

i The matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ is said to satisfy the RIP with parameters (K, δ), if for all $\mathcal{T} \subset \{1, \cdots, n\}$ such that $|\mathcal{T}| \leq K$ and for all $\boldsymbol{q} \in \mathbb{R}^{|\mathcal{T}|}$, it holds that

$$(1-\delta) \|\boldsymbol{q}\|_2^2 \leq \|\boldsymbol{A}_{\mathcal{T}} \boldsymbol{q}\|_2^2 \leq (1+\delta) \|\boldsymbol{q}\|_2^2.$$

The RIC δ_K is defined as the smallest constant δ for which the (K, δ)-RIP holds, i.e.,

$$\delta_K = \inf \left\{ \delta : \; (1-\delta) \|\boldsymbol{q}\|_2^2 \leq \|\boldsymbol{A}_{\mathcal{T}} \boldsymbol{q}\|_2^2 \leq (1+\delta) \|\boldsymbol{q}\|_2^2 \right.$$
$$\left. \forall\, |\mathcal{T}| \leq K, \; \forall \boldsymbol{q} \in \mathbb{R}^{|\mathcal{T}|} \right\}.$$

[3]

ii The matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ is said to satisfy the RIP with RIC $\delta_K \in (0, 1)$, if for all $\mathcal{T} \subset \{1, \cdots, n\}$ such that $|\mathcal{T}| \leq K$, it holds that

$$\sqrt{1-\delta_K} \leq \sigma_{\min}(\boldsymbol{A}_{\mathcal{T}}) \leq \sigma_{\max}(\boldsymbol{A}_{\mathcal{T}}) \leq \sqrt{1+\delta_K}.$$

[3]

(c)

i It is clear that

$$\left\| \begin{bmatrix} \boldsymbol{a}' \\ \boldsymbol{b}' \end{bmatrix} \right\|_2^2 = \sum_i (a_i')^2 + \sum_j (b_j')^2$$
$$= \|\boldsymbol{a}'\|_2^2 + \|\boldsymbol{b}'\|_2^2 = 2.$$

Similarly,

$$\left\| \begin{bmatrix} \boldsymbol{a}' \\ -\boldsymbol{b}' \end{bmatrix} \right\|_2^2 = 2.$$

[2]

ii As

$$\boldsymbol{x}' + \boldsymbol{y}' = [\boldsymbol{A}_{\mathcal{I}} \boldsymbol{A}_{\mathcal{J}}] \begin{bmatrix} \boldsymbol{a}' \\ \boldsymbol{b}' \end{bmatrix},$$

RIP implies

$$2\left(1-\delta_{2k}\right) \leq \left\|\boldsymbol{x}'+\boldsymbol{y}'\right\|_2^2 \leq 2\left(1+\delta_{2k}\right).$$

Similarly, it is also true that

$$2\left(1-\delta_{2k}\right) \leq \left\|\boldsymbol{x}'-\boldsymbol{y}'\right\|_2^2 \leq 2\left(1+\delta_{2k}\right).$$

[3]

iii It holds that

$$\left\langle \boldsymbol{x}', \boldsymbol{y}' \right\rangle \leq \frac{\left\|\boldsymbol{x}'+\boldsymbol{y}'\right\|_2^2 - \left\|\boldsymbol{x}'-\boldsymbol{y}'\right\|_2^2}{4} \leq \delta_{2k},$$

and

$$-\left\langle \boldsymbol{x}', \boldsymbol{y}' \right\rangle \leq \frac{\left\|\boldsymbol{x}'-\boldsymbol{y}'\right\|_2^2 - \left\|\boldsymbol{x}'+\boldsymbol{y}'\right\|_2^2}{4} \leq \delta_{2k}.$$

As a consequence,

$$\left|\left\langle \boldsymbol{x}', \boldsymbol{y}' \right\rangle\right| \leq \delta_{2k},$$

and

$$\left|\left\langle \boldsymbol{A}_{\mathcal{I}} \boldsymbol{a}, \boldsymbol{A}_{\mathcal{J}} \boldsymbol{b} \right\rangle\right| \leq \delta_{2k} \left\|\boldsymbol{a}\right\|_2 \left\|\boldsymbol{b}\right\|_2.$$

It concludes that

$$c = \delta_{2k}.$$

[3]

iv

$$\begin{aligned} \left\|\boldsymbol{A}_{\mathcal{I}}^T \boldsymbol{A}_{\mathcal{J}} \boldsymbol{b}\right\|_2 &= \sup_{\boldsymbol{a}:\, \|\boldsymbol{a}\|_2=1} \left|\boldsymbol{a}^T \left(\boldsymbol{A}_{\mathcal{I}}^T \boldsymbol{A}_{\mathcal{J}} \boldsymbol{b}\right)\right| \\ &= \sup_{\boldsymbol{a}:\, \|\boldsymbol{a}\|_2=1} \left|\left\langle \boldsymbol{A}_{\mathcal{I}} \boldsymbol{a}, \boldsymbol{A}_{\mathcal{J}} \boldsymbol{b} \right\rangle\right| \\ &\leq \delta_{2k} \left\|\boldsymbol{b}\right\|_2. \end{aligned}$$

[2]

(d)

i

$$\left\|\boldsymbol{A}_{\mathcal{J}}^T \boldsymbol{y}\right\|_2 = \left\|\boldsymbol{A}_{\mathcal{J}}^T \boldsymbol{A}_{\mathcal{I}} \boldsymbol{x}_{\mathcal{I}}\right\|_2 \leq \delta_{2k} \left\|\boldsymbol{x}\right\|_2.$$

[2]

ii

$$\begin{aligned}\left\|\boldsymbol{A}_{\mathcal{I}}^{T}\boldsymbol{y}\right\|_2 &= \left\|\boldsymbol{A}_{\mathcal{I}}^{T}\boldsymbol{A}_{\mathcal{I}}\boldsymbol{x}_{\mathcal{I}}\right\|_2 \\ &\geq (1-\delta_k)\left\|\boldsymbol{x}\right\|_2 \geq (1-\delta_{2k})\left\|\boldsymbol{x}\right\|_2 .\end{aligned}$$

[2]

iii In order to make $\left\|\boldsymbol{A}_{\mathcal{I}}^{T}\boldsymbol{y}\right\|_2 \geq \left\|\boldsymbol{A}_{\mathcal{J}}^{T}\boldsymbol{y}\right\|_2$, it is sufficient to have $1-\delta_{2k} \geq \delta_{2k}$, i.e., $\delta_{2k} \leq \frac{1}{2}$. [1]

iv

$$\begin{aligned}\left\|\boldsymbol{A}_{\mathcal{J}}^{\dagger}\boldsymbol{y}\right\|_2 &= \left\|\left(\boldsymbol{A}_{\mathcal{J}}^{T}\boldsymbol{A}_{\mathcal{J}}\right)^{-1}\boldsymbol{A}_{\mathcal{J}}^{T}\boldsymbol{A}_{\mathcal{I}}\boldsymbol{x}_{\mathcal{I}}\right\|_2 \\ &\leq \frac{1}{1-\delta_{2k}}\left\|\boldsymbol{A}_{\mathcal{J}}^{T}\boldsymbol{A}_{\mathcal{I}}\boldsymbol{x}_{\mathcal{I}}\right\|_2 \\ &\leq \frac{\delta_{2k}}{1-\delta_{2k}}\left\|\boldsymbol{x}\right\|_2 .\end{aligned}$$

At the same time,

$$\begin{aligned}\left\|\boldsymbol{A}_{\mathcal{I}}^{\dagger}\boldsymbol{y}\right\|_2 &= \left\|\left(\boldsymbol{A}_{\mathcal{I}}^{T}\boldsymbol{A}_{\mathcal{I}}\right)^{-1}\boldsymbol{A}_{\mathcal{I}}^{T}\boldsymbol{A}_{\mathcal{I}}\boldsymbol{x}_{\mathcal{I}}\right\|_2 \\ &= \left\|\boldsymbol{x}_{\mathcal{I}}\right\|_2 = \left\|\boldsymbol{x}\right\|_2 .\end{aligned}$$

In order to make $\left\|\boldsymbol{A}_{\mathcal{I}}^{\dagger}\boldsymbol{y}\right\|_2 \geq \left\|\boldsymbol{A}_{\mathcal{J}}^{\dagger}\boldsymbol{y}\right\|_2$, it is sufficient to have $1 \geq \frac{\delta_{2k}}{1-\delta_{2k}}$, i.e., $\delta_{2k} \leq \frac{1}{2}$. [2]

Solution to Problem 2.

(a)

i A set $\mathcal{S} \subset \mathbb{R}^n$ is convex if

$$\lambda \boldsymbol{x} + (1-\lambda)\,\boldsymbol{y} \in \mathcal{S},$$

for all $\boldsymbol{x}, \boldsymbol{y} \in \mathcal{S}$ and all $\lambda \in [0,1]$. [2]

ii A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if

$$f(\lambda \boldsymbol{x} + (1-\lambda)\,\boldsymbol{y}) \leq \lambda f(\boldsymbol{x}) + (1-\lambda)\, f(\boldsymbol{y})$$

for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and all $\lambda \in [0,1]$. [2]

(b)

i It is clear that $\nabla f(\boldsymbol{x}) \in \mathbb{R}^n$, $\nabla_{\boldsymbol{y}-\boldsymbol{x}} f(\boldsymbol{x}) \in \mathbb{R}$, and

$$\nabla_{\boldsymbol{y}-\boldsymbol{x}} f(\boldsymbol{x}) = \nabla f(\boldsymbol{x})^T (\boldsymbol{y}-\boldsymbol{x}).$$

[2]

ii By definition of convexity, for all $\lambda \in [0,1]$,

$$f((1-\lambda)\,\boldsymbol{x} + \lambda \boldsymbol{y}) \leq (1-\lambda)\, f(\boldsymbol{x}) + \lambda f(\boldsymbol{y}).$$

This implies

$$\lambda f(\boldsymbol{y}) \geq \lambda f(\boldsymbol{x}) + f(\boldsymbol{x} + \lambda(\boldsymbol{y}-\boldsymbol{x})) - f(\boldsymbol{x}),$$

or equivalently,

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \frac{f(\boldsymbol{x} + \lambda(\boldsymbol{y}-\boldsymbol{x})) - f(\boldsymbol{x})}{\lambda}.$$

Take the limit $\lambda \to 0$. One has

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T (\boldsymbol{y}-\boldsymbol{x}).$$

[2]

iii A vector $\boldsymbol{v} \in \mathbb{R}^n$ is a subgradient of $f(\boldsymbol{x})$ at a point $\boldsymbol{x}$ if

$$f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \boldsymbol{v}^T(\boldsymbol{y} - \boldsymbol{x}) \text{ for all } \boldsymbol{y} \in \mathbb{R}^n.$$

[2]

iv For $f(x) = |x|$, its subdifferential is given by

$$\partial f = \begin{cases} 1 & \text{if } x > 0, \\ [-1, 1] & \text{if } x = 0, \\ -1 & \text{if } x < 0. \end{cases}$$

[3]

(c)

i As proved in (2.3), it holds that for all $\boldsymbol{y} \in \mathbb{R}^n$,

$$\begin{aligned} f(\boldsymbol{y}) &\geq f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^T(\boldsymbol{y} - \boldsymbol{x}) \\ &= f(\boldsymbol{x}), \end{aligned}$$

where the equality follows from the fact that $\nabla f(\boldsymbol{x}) = \mathbf{0}$. Hence $\boldsymbol{x}$ is a global minimiser of f. [2]

ii By the definition of subgradient, it holds that for all $\boldsymbol{y} \in \mathbb{R}^n$,

$$\begin{aligned} f(\boldsymbol{y}) &\geq f(\boldsymbol{x}) + \mathbf{0}^T(\boldsymbol{y} - \boldsymbol{x}) \\ &= f(\boldsymbol{x}). \end{aligned}$$

Hence $\boldsymbol{x}$ is a global minimiser of f. [2]

iii The soft-thresholding function is of the form

$$x^* = \eta(z; \lambda) = \begin{cases} z - \lambda & \text{if } z \geq \lambda, \\ 0 & \text{if } -\lambda < z < \lambda, \\ z + \lambda & \text{if } z \leq -\lambda. \end{cases}$$

[2]

iv The IST algorithm is an iterative algorithm where in the kth iteration the variable $\boldsymbol{x}^k$ is updated by

$$\boldsymbol{x}^k = \eta\left(\boldsymbol{x}^{k-1} + t_k \boldsymbol{A}^T\left(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}^{k-1}\right); \lambda t_k\right),$$

and $t_k > 0$ is an appropriately chosen step size. [3]

v Define the soft thresholding function $\eta_\sigma(\boldsymbol{Z}; \lambda)$ as

$$\boldsymbol{X} = \eta_\sigma(\boldsymbol{Z}; \lambda) = \sum_i \boldsymbol{u}_i \eta(\sigma_i; \lambda) \boldsymbol{v}_i^T,$$

where σ_i is the ith singular value of $\boldsymbol{Z}$, $\boldsymbol{u}_i$ and $\boldsymbol{v}_i$ are the corresponding singular vectors. The IST algorithm to solve the low-rank matrix recovery problem is an iterative algorithm where in the kth iteration the matrix $\boldsymbol{X}^k$ is updated by

$$\boldsymbol{X}^k = \eta_\sigma\left(\boldsymbol{X}^{k-1} + t_k \mathcal{A}^*\left(\boldsymbol{y} - \mathcal{A}\left(\boldsymbol{X}^{k-1}\right)\right); \lambda t_k\right),$$

where $\mathcal{A}^*$ is the adjoint operator (transpose) of $\mathcal{A}$ and $t_k > 0$ is an appropriately chosen step size. [3]

Solution to Problem 3.

(a)

i The standard form of a convex optimisation problem is

$$\begin{aligned} &\min_{\boldsymbol{x}} f(\boldsymbol{x}) \\ &\text{subject to } h_i(\boldsymbol{x}) \leq 0, \quad i \in \{1, \cdots, m\} \\ &\qquad \ell_j(\boldsymbol{x}) = 0, \quad j \in \{1, \cdots, p\} \end{aligned}$$

where $f(\boldsymbol{x})$, $h_i(\boldsymbol{x})$ are convex and ℓ_j is affine. [2]

ii The corresponding Lagrangian is given by

$$L(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{v}) = f(\boldsymbol{x}) + \sum_{i=1}^{m} u_i h_i(\boldsymbol{x}) + \sum_{j=1}^{p} v_j \ell_j(\boldsymbol{x})$$

where $\boldsymbol{u} \in \mathbb{R}^m$, $\boldsymbol{v} \in \mathbb{R}^p$, and $\boldsymbol{u} \geq \boldsymbol{0}$. [2]

iii The Lagrange dual function is given as

$$g(\boldsymbol{u}, \boldsymbol{v}) = \min_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{v}).$$

And the Lagrange dual problem is stated as

$$\begin{aligned} &\max_{\boldsymbol{u} \in \mathbb{R}^m,\, \boldsymbol{v} \in \mathbb{R}^p} g(\boldsymbol{u}, \boldsymbol{v}) \\ &\text{subject to } \boldsymbol{u} \geq \boldsymbol{0}. \end{aligned}$$

[2]

iv The KKT conditions are given as

- $\boldsymbol{0} \in \partial f(\boldsymbol{x}) + \sum_{i=1}^{m} u_i \partial h_i(\boldsymbol{x}) + \sum_{j=1}^{p} v_j \partial \ell_j(\boldsymbol{x})$ [1/4]
- $u_i h_i(\boldsymbol{x}) = 0$ for all $i \in \{1, \cdots, m\}$ [1/4]
- $h_i(\boldsymbol{x}) \leq 0$, $\ell_j(\boldsymbol{x}) = 0$, and $u_i \geq 0$ for all $i \in \{1, \cdots, m\}$ and $j \in \{1, \cdots, p\}$. [2/4]

(b)

i

$$\begin{aligned} L_\rho(\boldsymbol{x}, \boldsymbol{z}, \boldsymbol{v}) = f(\boldsymbol{x}) + g(\boldsymbol{z}) + \frac{\rho}{2} \|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\|_2^2 \\ + \boldsymbol{v}^T (\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}). \end{aligned}$$

[1]

ii

- The variable $\boldsymbol{x}$ is updated via

$$\begin{aligned}\boldsymbol{x}^{k+1} &= \arg\min_{\boldsymbol{x}} L_\rho\left(\boldsymbol{x}, \boldsymbol{z}^k, \boldsymbol{v}^k\right) \\ &= \arg\min_{\boldsymbol{x}} f\left(\boldsymbol{x}\right) + \frac{\rho}{2}\left\|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z}^k - \boldsymbol{c}\right\|_2^2 \\ &\quad + \left(\boldsymbol{v}^k\right)^T\left(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z}^k - \boldsymbol{c}\right).\end{aligned}$$

[2/4]

- The variable $\boldsymbol{z}$ is updated via

$$\begin{aligned}\boldsymbol{z}^{k+1} &= \arg\min_{\boldsymbol{z}} L_\rho\left(\boldsymbol{x}^{k+1}, \boldsymbol{z}, \boldsymbol{v}^k\right) \\ &= \arg\min_{\boldsymbol{z}} g\left(\boldsymbol{z}\right) + \frac{\rho}{2}\left\|\boldsymbol{A}\boldsymbol{x}^{k+1} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\right\|_2^2 \\ &\quad + \left(\boldsymbol{v}^k\right)^T\left(\boldsymbol{A}\boldsymbol{x}^{k+1} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c}\right).\end{aligned}$$

[2/4]

iii By replacing $\boldsymbol{v}$ with $\boldsymbol{w} = \frac{1}{\rho}\boldsymbol{v}$, the three steps in each iteration of ADMM can be simplified to

$$\begin{aligned}\boldsymbol{x}^{k+1} &= \arg\min_{\boldsymbol{x}} f\left(\boldsymbol{x}\right) + \frac{\rho}{2}\left\|\boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{z}^k - \boldsymbol{c} + \boldsymbol{w}^k\right\|_2^2 \\ \boldsymbol{z}^{k+1} &= \arg\min_{\boldsymbol{z}} g\left(\boldsymbol{z}\right) + \frac{\rho}{2}\left\|\boldsymbol{A}\boldsymbol{x}^{k+1} + \boldsymbol{B}\boldsymbol{z} - \boldsymbol{c} + \boldsymbol{w}^k\right\|_2^2 \\ \boldsymbol{w}^{k+1} &= \boldsymbol{w}^k + \boldsymbol{A}\boldsymbol{x}^{k+1} + \boldsymbol{B}\boldsymbol{z}^{k+1} - \boldsymbol{c}.\end{aligned}$$

[2]

(c)

i The ADMM formulation is given by

$$\begin{aligned}&\min_{\boldsymbol{x},\boldsymbol{z}} \frac{1}{2}\left\|\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}\right\|_2^2 + \lambda\left\|\boldsymbol{z}\right\|_1 \\ &\text{subject to } \boldsymbol{x} - \boldsymbol{z} = \boldsymbol{0}.\end{aligned}$$

[2/3]

The scaled form of ADMM iterations are given by

$$\begin{aligned}
\boldsymbol{x}^{k+1} &= \arg\min_{\boldsymbol{x}} \frac{1}{2}\|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2^2 + \frac{\rho}{2}\left\|\boldsymbol{x}-\boldsymbol{z}^k+\boldsymbol{w}^k\right\|_2^2 \\
\boldsymbol{z}^{k+1} &= \arg\min_{\boldsymbol{z}} \lambda\|\boldsymbol{z}\|_1 + \frac{\rho}{2}\left\|\boldsymbol{x}^{k+1}-\boldsymbol{z}+\boldsymbol{w}^k\right\|_2^2 \\
\boldsymbol{w}^{k+1} &= \boldsymbol{w}^k + \boldsymbol{x}^{k+1} - \boldsymbol{z}^{k+1}.
\end{aligned}$$

[1/3]

ii The ADMM formulation is given by

$$\begin{aligned}
&\min_{\boldsymbol{x},\boldsymbol{z}} \frac{1}{2}\|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2^2 + g(\boldsymbol{z}) \\
&\text{subject to } \underbrace{\begin{bmatrix}\boldsymbol{I}\\ \boldsymbol{B}\end{bmatrix}}_{\boldsymbol{B}'}\boldsymbol{x} + \underbrace{\begin{bmatrix}-\boldsymbol{I} & \\ & \boldsymbol{I}\end{bmatrix}}_{\boldsymbol{D}'}\underbrace{\begin{bmatrix}\boldsymbol{z}_1\\ \boldsymbol{z}_2\end{bmatrix}}_{\boldsymbol{z}} = \boldsymbol{0},
\end{aligned}$$

where

$$\begin{aligned}
g(\boldsymbol{z}) &= \lambda\|\boldsymbol{z}_1\|_1 + \mathbb{1}_{\geq 0}(\boldsymbol{z}_2) \\
\mathbb{1}_{\geq 0}(z) &= \begin{cases}\infty & \text{if } z < 0 \\ 0 & \text{if } z \geq 0\end{cases}.
\end{aligned}$$

[3/5]

The scaled form of ADMM iterations are given by

$$\begin{aligned}
\boldsymbol{x}^{k+1} &= \arg\min_{\boldsymbol{x}} \frac{1}{2}\|\boldsymbol{y}-\boldsymbol{A}\boldsymbol{x}\|_2^2 + \frac{\rho}{2}\left\|\boldsymbol{B}'\boldsymbol{x}+\boldsymbol{D}'\boldsymbol{z}^k+\boldsymbol{w}^k\right\|_2^2 \\
\boldsymbol{z}^{k+1} &= \arg\min_{\boldsymbol{z}_1,\boldsymbol{z}_2} \lambda\|\boldsymbol{z}_1\|_1 + \frac{\rho}{2}\left\|\boldsymbol{x}^{k+1}-\boldsymbol{z}_1+\boldsymbol{w}_1^k\right\|_2^2 \\
&\qquad + \mathbb{1}_{\geq 0}(\boldsymbol{z}_2) + \frac{\rho}{2}\left\|\boldsymbol{B}\boldsymbol{x}^{k+1}+\boldsymbol{z}_2+\boldsymbol{w}_2^k\right\|_2^2 \\
\boldsymbol{w}^{k+1} &= \boldsymbol{w}^k + \boldsymbol{B}'\boldsymbol{x}^{k+1} + \boldsymbol{D}'\boldsymbol{z}^{k+1}.
\end{aligned}$$

[2/5]